

Education

Bachelor of Science in Computer and Information Science; Bachelor of Arts in Chinese

The Ohio State University: Columbus, OH

Specialization: *Information & Computational Assurance*

August 2022 – December 2026

- GPA: 3.86 (Dean's List all completed terms) - student member of ACM (htrowbridge@acm.org)
- Relevant Coursework: Java, Operating Systems, Web Applications, Data Structures and Algorithms, Information Security

Work Experience

IT Intern – Enterprise Cybersecurity Policy

Chicago, IL

Abbott Laboratories

May 2026 – August 2026

- Evaluated 3 cloud-integrated AI transcription models for a call-center dashboard by engineering 4 novel metrics to optimize data capture and triage; proposed adoption of a more performant alternative projected to reduce operating costs by 75%
- Spearheaded the end-to-end proof-of-concept for an internal cyber policy AI assistant, delivering 40 pages of reproducible engineering documentation
 - Authored a structured 50-question evaluation test set to audit LLM outputs, resulting in an 86% accuracy rate as validated by 9 subject matter experts
- Accelerated global control-objective modernization by designing a proposal to consolidate legacy controls, reducing redundancies, easing adoption, and maximizing enterprise auditability
- Executed a preliminary gap analysis mapping internal cybersecurity controls against NIST 800-218 secure software development lifecycle (SSDL) standards
- Formulated a global training strategy and delivery platform to scale cyber policy tooling adoption and drive compliance across international business units
- Engineered an automated financial data pipeline that aggregates disjointed cost-center statements into a unified budget dashboard, delivering critical visibility to leadership and eliminating 10 hours of manual monthly labor

Undergraduate Research Assistant (Industrial Cybersecurity Lab)

Columbus, OH

Center for Design and Manufacturing Excellence

May 2025 – Present

- Co-led development of an [interactive web app](#) to help small manufacturers achieve [CMMC 2.0 cybersecurity compliance](#) needed for DoD contracts
- Developed a full-stack web app using container technologies, a custom database with role-based access controls (RBAC), and Python APIs for backend functionality; managed with Agile/Scrum and frequent client meetings
- Ran blockchain (hyperledger fabric) pruning and performance-measuring experiments

Software and Data Engineer Intern

Washington, D.C. (Remote)

GovRat

September 2025 – May 2026

- Developed Python-based automation to ingest and structure federal contracting data
- Ensured data reliability and consistency for in-house LLM (Large Language Model) training, directly enhancing the platform's ability to draft compliance-ready government-contract proposals

Classroom and Digital Union Support Technician

Columbus, OH

The Ohio State University Office of Technology and Digital Innovation

March 2025 – September 2025

- Resolved mission-critical IT/AV failures in classrooms, ensuring uninterrupted instruction for hundreds of students weekly
- Delivered faculty training and troubleshooting support that boosted classroom tech uptime and reduced repeat service calls

Accounting Intern

Chicago, IL

Jeffrey W. Krol & Associates, Ltd.

June 2023 – August 2023; May 2024 – July 2024

- Migrated client data from QuickBooks Online to Desktop, creating standardized documentation that colleagues reused
- Built digital archival system and Excel dashboards, reducing reporting turnaround time and improving investment tracking

Technical Projects and Research

U.S. Cyber Command Research Competition: Cyber RECon 2026's Winning Paper

Fort Meade, MD

Semantic Shields: Automating Critical Infrastructure Defense via NLP-Driven Ransomware Profiling

April 2026

- [Led the development, writing, research, submission, and publication](#) of a novel research paper (as first author), supported by [my lab](#) working with a cross-functional team
- Won all three categories of US Cyber Command's 2026 Cyber RECon research competition - overall best research, best research paper in the analyst category, and best presentation (poster and slide-deck)
- Argued for the adoption of NLP (Natural Language Processing) to protect US critical infrastructure

Debian Package Mirrors

June 2025 – February 2026

- Deployed Debian mirrors on FreeBSD (complete) and Debian with NGINX, provided reliable package access for users
- Automated synchronization with official repositories, ensuring updated packages with minimal downtime

Henry Codes - Personal Portfolio (Live-Hosted) | [GitHub](#)

January 2025 – Present

- Designed and self-hosted personal portfolio site (on Debian, NGINX, PHP) with Cloudflare -- secure, reproducible, Docker-containerized

Certifications and Qualifications

- Centrix™ Power User | Armis | Centrix™ Power User Learning Path (7.5 CPE)

- **Security*** (in-progress) | CompTIA

Leadership and Extracurricular Activities

- **Vice President, [Cybersecurity Club](#)** | Nov 2022 – Present | Organized events (BuckeyeCTF, Ohio LinuxFest collab)
- **Vice President, [Open Source Club](#)** | Sep 2024 – Present | Advocated for Linux and open-source tools
- **Member, [History Club](#)** | Oct 2022 – Present | Learned history; presented on Chinese three-kingdoms periods and other topics
- **Consulting Member (Former President), [Buckeye Table Tennis Club](#)** | Aug 2022 – Present | Participant in club practice

Skills

- **Professional:** Adaptability, Cross-Functional Collaboration, Client Communication, Project Management, Technical Documentation, Change Management, Agile/Scrum, English (Fluent), Chinese (Professional Working Proficiency)
- **Infrastructure & Security:** Governance, Risk, and Compliance; AI Strategy; AI Model Evaluation; Cyber Risk Management; Process Automation; CMMC (NIST 800-171r3), Linux (Debian, RHEL), FreeBSD, Docker, Podman, Git, NGINX, SSH, RDP, UFW, Firewalls, Web servers, Cloudflare, Tailscale, Zero Trust, Principle of Least Privilege, Access Control Lists, Role-Based Access Controls
- **Backend Development:** Python (FastAPI, Flask, SQLAlchemy, Alembic, Pandas, Plotly, REST APIs), Ruby (Ruby on Rails), Java, C#, C, SQL (PostgreSQL, SQLite)
- **Frontend Development:** HTML, CSS, JavaScript, Jinja, TailwindCSS